TEXORA · LECTURE SHEET

Solid Waste Management and Costing

What a textile factory's solid waste consists of, how it is collected and segregated, the treatment and re-use alternatives for each stream, how water use is cut by process control and recycle — and how the cost of waste treatment is determined.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 7 · LEVEL 4 TERM 2

1. Composition of textile solid waste

Solid-waste management starts with an audit: what arises, where, how much, and how hazardous. For a composite knit dyeing and garment factory the streams are typically of the following kind (shares are illustrative of such an audit, and every factory must measure its own):

STREAM	SOURCE	ILLUSTRATIVE SHARE	CHARACTER
ETP sludge	Effluent treatment (chemical + biological)	35 %	Wet, contains removed dyes/chemicals; the problem stream
Fabric and yarn waste	Cutting, sewing, knitting rejects	20 %	Clean, saleable for recycling if kept uncontaminated
Boiler ash	Solid-fuel boilers	10 %	Inert mineral residue
Metal and miscellaneous scrap	Maintenance, worn parts	10 %	Saleable scrap
Packaging waste	Polythene, cartons, cones, drums (non-chemical)	12 %	Mostly recyclable
Office and domestic refuse	Canteen, offices, dormitories	8 %	Municipal-type waste
Chemical containers with residues	Dye/chemical stores	5 %	Hazardous until decontaminated

Classification drives everything after it: **inert** waste (ash), **biodegradable** waste (food, some sludge organics), **recyclable** waste (fabric, packaging, metal), and **hazardous** waste (chemical containers,

sludge failing toxicity tests, waste oils, contaminated rags) each have their own legal and practical route.

2. Collection systems and segregation

The cheapest tonne of waste to manage is the one that was never mixed. **Segregation at source** — colour-coded bins and skips at the point of generation (fabric waste by fibre type and colour, packaging, food waste, chemical containers, general refuse) — keeps recyclable streams clean and saleable and keeps hazardous streams small. The **collection system** then moves each stream on a fixed schedule to a central, bunded, signed waste yard: covered storage for sludge (rain turns stored sludge back into liquid effluent), sealed containers for hazardous items with labels and a manifest record, and baling for fabric and carton waste to cut volume and raise resale value. Housekeeping rules — no burning of waste on site, no mixing of chemical containers into general waste, spill kits at transfer points — are part of the system, enforced by the same audit routine that weighs and records each outgoing load.

3. Treatment and re-use alternatives, stream by stream

STREAM	ALTERNATIVES IN ORDER OF PREFERENCE
Fabric / yarn waste	Re-use as cutting-room seconds → mechanical recycling (shoddy fibre, yarn spinning) → wiping cloths and filling → energy recovery → landfill (last)
ETP sludge	Dewater (filter press) and dry (drying beds / solar) to cut mass → tested, controlled uses (e.g. as a minor raw-material fraction in bricks, where leaching tests pass) → engineered/secure landfill. Never open dumping or use on food land without toxicity clearance
Chemical containers	Return to supplier (refill schemes) → triple-rinse and puncture, then recycle as plastic/metal → hazardous-waste handler if residues remain
Boiler ash	Brick and block making, road sub-base and land filling under conditions preventing dust and leaching
Packaging	Re-use (cones, pallets) → sale to recyclers by grade
Organic / canteen waste	Composting or biogas; the digestate returns as soil conditioner

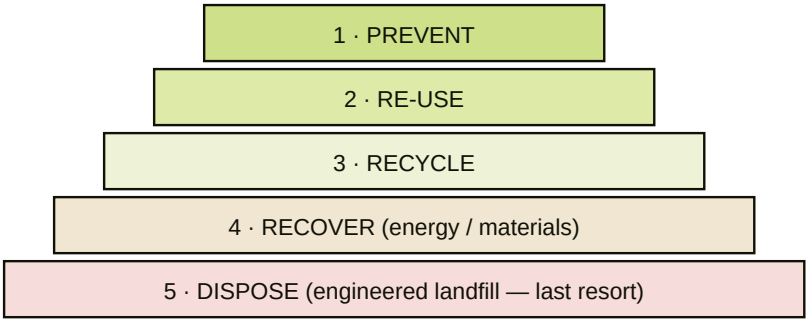


Figure 1 — The waste hierarchy. Management effort and money are spent from the top down: a tonne prevented is worth more than a tonne recycled, and disposal is what remains when every better option is exhausted.

Two principles govern the choice. The **waste hierarchy** above: prevention outranks re-use, which outranks recycling, then recovery, with disposal last — a saleable recycling stream beats an energy-recovery contract, which beats landfill. And **liability follows the waste**: the generator remains responsible for what its contractor does with sludge and hazardous waste, so destinations are audited, not assumed.

4. Reducing wastewater at source: automatic process control, recycle and filtration

Solid-waste thinking applies to water as well, and the syllabus pairs them deliberately. **Automatic process control** cuts effluent volume and load before treatment: automatic chemical dispensing doses what the recipe needs (not a operator's guess), temperature and pH controllers hold baths at setpoints, and right-first-time dyeing (Topic 3) eliminates the reprocessing that doubles a batch's water and chemical footprint. Low-liquor-ratio machines and counter-current rinsing cut volume structurally. **Recycle and filtration** then return water to service: cooling water and steam condensate are inherently clean loops; final rinse water can serve as the next batch's first rinse; and membrane filtration (ultrafiltration, reverse osmosis) of treated effluent produces process-grade water, with the rejected concentrate returned to treatment. **Wastewater analysis** — the routine measurement of flow, pH, BOD, COD, TDS, colour and stream-specific parameters — is what makes all of this manageable: a factory that cannot state its per-kilogram water and load figures cannot improve them, and the analysis log is also its evidence of compliance (Topic 8).

5. Determining the cost of waste treatment

Waste treatment is a production cost and must be computed like one. The method: collect all **operating costs** of the ETP and waste system for a period (chemicals, power, labour, sludge handling and disposal, maintenance, laboratory and compliance costs), annualise the **capital cost** (depreciation on plant, interest) where full costing is wanted, then divide by the throughput of the same period —

expressed per cubic metre treated and per kilogram of fabric processed, the two units managers actually use.

WORKED EXAMPLE — COST OF EFFLUENT TREATMENT

A factory treats 600 m³/day of effluent and processes 450,000 kg of fabric in a 30-day month (treated volume = 600 × 30 = 18,000 m³). Monthly ETP operating costs:

Chemicals Tk 2,70,000 · Power Tk 1,80,000 · Labour Tk 90,000 · Sludge disposal Tk 60,000 · Maintenance Tk 50,000

Total = 2,70,000 + 1,80,000 + 90,000 + 60,000 + 50,000 = **Tk 6,50,000 per month**.

Cost per cubic metre = 6,50,000 ÷ 18,000 = **Tk 36.11 per m³**.

Cost per kilogram of fabric = 6,50,000 ÷ 450,000 = **Tk 1.44 per kg**.

Uses: the per-kg figure enters product costing and buyer price discussions; the breakdown shows chemicals are the largest single item (≈ 42 % of the total), so a coagulant-dose optimisation trial attacks the right cost; and the figure is the baseline against which water-recycle investments (which cut both intake and discharge volumes) are appraised.

Cost determination pitfalls mirror those of effluent design: omitting sludge disposal (the cost reappears as an illegal-dumping risk), ignoring capital charges when comparing against a common-ETP tariff, and averaging away the fact that a difficult shade programme costs more to treat than a light one. Where a central effluent treatment plant (CETP) serves an industrial estate, the same computation is done by the operator and recovered as a tariff — the factory then compares tariff × volume against its own full cost per cubic metre on equal terms.

Glossary

Segregation at source — separating waste streams where they arise, before mixing. **Waste hierarchy** — priority order: prevent, re-use, recycle, recover, dispose. **Dewatering** — mechanical removal of water from sludge (e.g. filter press) before drying or disposal. **Hazardous waste** — waste dangerous by toxicity, flammability, corrosivity or reactivity, needing special handling. **CETP** — central (common) effluent treatment plant serving several factories. **Counter-current rinsing** — arranging rinse flow opposite to fabric flow so the cleanest water meets the cleanest fabric. **Manifest** — the tracking record accompanying a hazardous-waste consignment. **Leaching** — dissolution of contaminants out of waste by percolating water.

Key points

• Audit first: the streams are sludge, fabric waste, ash, packaging, scrap, refuse and chemical containers — each with its own route, hazardous ones strictly separated.

• Segregation at source, scheduled collection and a proper waste yard cost little and determine the value (or the hazard) of everything downstream.
• Re-use and recycling beat recovery, which beats disposal; the generator keeps liability for its waste wherever it goes.
• Automatic process control, low-liquor processing, recycle and filtration cut water and load before treatment; routine wastewater analysis supplies the numbers.
• Treatment cost is determined by summing period costs and dividing by throughput: Tk 6,50,000/month ÷ 18,000 m^3 = Tk 36.11/m^3, or Tk 1.44 per kg of fabric in the worked example.

Practice

1. A factory's monthly ETP operating cost is Tk 5,85,000. It treats 450 m^3/day over a 30-day month and processes 3,90,000 kg of fabric. Find the treatment cost per m^3 and per kg.
2. Why does mixing clean fabric cuttings with general refuse destroy value, and what collection arrangement prevents it?
3. List, in hierarchy order, the management options for ETP sludge, and state the test concern that gates any use of sludge in bricks or on land.
4. Name three ways automatic process control reduces wastewater volume or load in a dyehouse, explaining the mechanism of each.
5. A buyer offers to take a factory's sludge away free of charge. What checks should the factory still make, and why?

Answer key

1. Monthly volume = 450 × 30 = 13,500 m^3. Cost per m^3 = 5,85,000 ÷ 13,500 = **Tk 43.33/m^3**. Cost per kg = 5,85,000 ÷ 3,90,000 = **Tk 1.50/kg**.
2. Clean cuttings are a saleable recycling raw material graded by fibre and colour; mixed with wet, dirty refuse they become contaminated, unsorted waste whose only route is paid disposal — value destroyed by mixing. Prevention: segregation at source with labelled colour-coded bins/skips at the cutting tables, scheduled separate collection, and baling/storage apart from general waste.
3. Hierarchy for sludge: (1) prevention/reduction — better treatment chemistry and source control reducing sludge generation; (2) dewatering and drying to cut mass; (3) controlled re-use/recovery — e.g. a minor fraction in bricks where tests pass; (4) engineered/secure landfill disposal. Open dumping is not an option at any level. Gating concern: leachability/toxicity — whether heavy metals and residual chemicals leach out in service or in the ground, shown by leaching (and where relevant, toxicity) tests.
4. Any three: automatic dispensing — doses exact recipe quantities, removing over-dosing that raises COD and sludge; temperature/pH control with right-first-time dyeing — eliminates reprocessing,

which would repeat the whole bath's water and chemicals; low-liquor-ratio machines and counter-current rinsing — structurally less water per kg for the same washing effect; level/flow control on rinses — stops overflow running-to-drain losses. (Mechanism must be explained for each chosen.)

5. Check the destination and method: where the sludge will go, whether the taker is authorised/equipped, what documentation (manifest/receipts) will evidence the transfer, and audit or visit the site. Reason: legal and reputational liability for the waste remains with the generator — "free removal" that ends in open dumping into a river is still the factory's offence and its headline.

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